STEAMROOT · DESIGN PRECIS

SteamRoot design precis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	STR-PRC-001	Version	0.8
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
Source	docs/02-concept.md @ 276b380		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: gas-pass baffles in the bank box and economizer (STR-DEC-001 S1, STEM check S6 and S7) with the flue openings and chimney moved; figures from STR-CAL-001 v0.5; requirement IDs from STR-PRS-001; concept options (STR-OPT-001) and STEM check (STR-STM-001) linked; after the independent review: chimney outlet 3.2 m (S3), latching pump trip (S4), 316L economizer baffles, efficiency 66.1 %
0.7	2026-10-02	Amish Chadha	Approved decisions carried into the design (STR-CAL-001 v0.4): evaporator bank and secondary air damper added, firebox 960 mm with 10 coil turns, seal pot overflow with loop seal (seal limit 0.084 bar), ballast-rated hood handles; figures and requirement status updated
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02: authority named, first co-design partner route, seal pot overflow and freezing rule, hoods on a second vehicle, evaporator

			bank study scope, hood ballast from TRL 4 tests
0.5	2026-10-01	Amish Chadha	Figures and description updated for the constructable design (STR-DDR-003, STR-CAL-001 v0.3); cost against the value-engineering target; build plan and decisions register linked
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update; record Amish's 2026-09-25 decisions; replace first-order numbers with STR-CAL-001 results; water-seal pot, two hoods and alarms added; regulatory ruling marked as an open prerequisite
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

SteamRoot design precis

SteamRoot is a towable, wood-fired steam generator that never holds pressure. A pump pushes water once through an economizer, an evaporator bank in the flue and a large-bore stainless coil above the fire, gas-pass baffles make the flue gas cross the bank and economizer tubes, the steam flows at close to atmospheric pressure through a hose to one of two insulated hoods pressed onto the soil, and primary and secondary air dampers hold the excess air near 1.5. The TRL 3 calculation (STR-CAL-001 v0.4) confirms the core safety idea on paper: 30 kg/h of steam with 7.3 L of water in the heated section and a header pressure of about 0.03 bar. With the evaporator bank and dampers decided on 2026-10-02 and the gas-pass baffles added on 2026-10-24 (STR-DEC-001 S1), fuel-to-steam efficiency is about 66 % (target 65 %, at risk: met only with the dampers set and the 3.2 m chimney outlet, and inside its uncertainty) and wood use 4.36 kg per m² at 15 cm (target 4, not met; a restatement to 4.5 is proposed). It treats about 1.9 m²/h at 15 cm with two hoods and weighs 509 kg as towed with the tank drained and both hoods aboard (over the 500 kg target). Value-engineering target: USD 2,200. Estimated cost of the constructable design: USD 2,720 (USD 520 over the target). The STEM check is in docs/01b-stem-check.md and the concept options, this design (A) and a vented fire-tube kettle (B), in docs/02a-concept-options.md; A and B score within 10 %, so the choice is with Amish. The design was made constructable on 2026-10-01 (STR-DDR-003, accepted by Amish on 2026-10-02); the prototype build plan is docs/05-build-plan.md and open decisions are in docs/06-design-decisions.md.

> **Safety:** SteamRoot is a concept for review, not a design to build. It combines open fire, carbon monoxide, and steam that causes severe burns in under a second. Local boiler and pressure vessel rules may apply even at low pressure. **A written ruling from the local boiler authority is an open prerequisite:** Amish approved getting it before TRL 3 detail work, and it has not been obtained. See the Safety section before any further work.



Figure 1. TRL 3 model in the working position: trailer, generator and two hoods on a bed, next to a 1.75 m person.

How it works

1. **Feed.** A 12 V diaphragm pump draws water from a 125 L tank and meters 0.5 L/min (30 kg/h) into the economizer at a fixed rate. A float switch warns when the tank runs low.
2. **Preheat and start to boil.** The economizer coil sits at the top of the flue and warms the feed water from about 15 °C to about 65 °C. The water then passes down through an evaporator bank, a serpentine of 15.88 mm stainless tube in a lined box on the firebox roof, where the hotter gas leaving the firebox brings it to the boil. A stainless baffle between the bank's two layers, and two more between the economizer's three layers, make the gas cross each layer of tubes in turn; the gas enters each box at the front and the chimney stands at the back of the economizer.
3. **Boil.** The warm water enters the bottom of a monotube coil that sits above the fire bed inside the fiber-lined firebox, on three brackets, with the door, grate and the two air inlets below it: primary air under the grate and secondary air through a slot in the door above the fire bed. It boils as it rises through 10 turns and leaves the top of the coil as steam. The coil holds about 5 L of water. A thermocouple on the coil outlet sounds an alarm if the outlet overheats, which is the sign of a coil running dry.
4. **Vent and separate.** The coil discharges into a steam header 1.55 m above the ground. The header is always open to the atmosphere through a water-seal pot: a dip leg reaches 890 mm below the water line. An overflow at the

static water mark, piped through a 350 mm loop seal down to ground level at the back of the trailer away from the operator's side (decided by Amish, 2026-10-02), stops the pot being overfilled and lets the water the dip leg displaces run away. Steam can therefore build only about 0.084 bar gauge (890 mm of water) before it bubbles through the seal and out of a vent pipe that ends 2.3 m above the ground, above head height. The seal cannot stick or be adjusted, and it is never valved off. A certified relief valve on the header is the second line of protection.

5. **Deliver.** Steam flows through a 6 m, 25 mm steam hose to a hood. A three-way diverter at the header sends steam to the hose or to the vent and has no closed position.
6. **Pasteurize.** Each hood is a 1.2 x 1.0 m insulated open-bottom pan with a 60 mm soil skirt. Steam from a perforated manifold condenses in the soil and drives a hot front downward. While one hood heats, the other holds its soil at temperature. To change over, the operator diverts steam to the vent, moves the hose coupling to the other hood and diverts steam back.

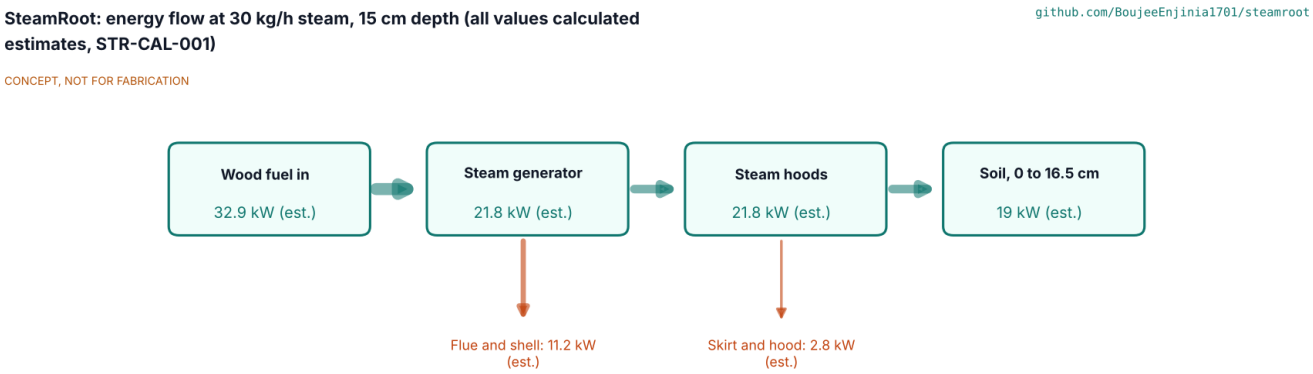


Figure 2. Energy flow at the design point of 30 kg/h of steam and 15 cm depth. All values are calculated estimates from STR-CAL-001.

Main components

Item numbers match the exploded view, the general arrangement drawing STR-DWG-002 and `bom/bom.csv`. Items 13 to 16 are not drawn.

Table 1. Main components.

#	Component	Choice	Notes
1	Trailer frame and drawbar	Used single-axle garden or utility trailer, deck 2.0 x 1.2 m	Rated 750 kg or more
2	Wheels	Trailer wheels, about 460 mm diameter	Included with the trailer
3	Feed water tank	125 L HDPE drum with strainer and sight tube	4.2 h of steaming
4	Feed pump	12 V diaphragm pump with needle valve and flow meter	Its 3 bar shut-off pressure also caps coil inlet pressure if the coil blocks
5	Firebox	700 x 600 x 960 mm steel box on two skids, bolted roof, 50 mm ceramic fiber lining, grate on a stand, latched door, primary air damper under the door and secondary air slide on the door	About 32 kW firing. A castable lining would add about 270 kg
6	Monotube steam coil	316 stainless tube, 25.4 mm OD x 1.65 mm wall, 13.2 m in 10 turns on 420 mm	Decided by Amish, 2026-09-25; one turn fewer since 2026-10-02 to keep R7
22	Evaporator bank	6.4 m of 15.88 mm x 1.24 mm 316 stainless tube in two layers, in a lined 700 x 600 x 170 mm box between the roof and the economizer, with a 1.5 mm stainless gas-pass baffle between the layers	Decided 2026-10-02; baffle added 2026-10-24 (STR-DEC-001 S1); about 66 % with the dampers set
7	Flue-gas economizer	7.3 m of 12.7 mm stainless tube in three layers in a box on the bank box, two 1 mm 316L stainless gas-pass baffles, condensate drain	Flue condensate is acidic (dew point about 51 °C); baffles added 2026-10-24, 316L after the review
8	Chimney and spark arrestor	150 mm flue at the back of the economizer lid, outlet 3.2 m above ground, 6 mm mesh	Keeps flue gas above head height; raised from 2.4 m for draft (STR-DEC-001 S3)
9	Steam header and water-seal vent	DN50 header at 1.55 m on a post, 114 mm seal pot with an 890 mm dip leg, an overflow at the static water mark through a loop seal to the ground behind the trailer, vent to 2.3 m, three-way diverter with a vent line, quick coupling	Primary pressure limit. Decided by Amish, 2026-09-25
10	Certified relief valve	ASME Section IV steam safety valve, 15 psi (1.03 bar) set, 3/4 in, rated 170 kg/h (375 lb/h)	Secondary protection only; sized for the dry-coil refeed flash (decided by Amish, 2026-09-25)
11	Steam hose	EPDM saturated-steam hose, 25 mm bore, 6 m, ground-joint couplings, whip checks	Steam rated, not hot-water rated
12	Steam hoods (2)	1.2 x 1.0 x 0.25 m aluminum double-skin pans, 40 mm mineral wool, soil skirt, steel handles on foot plates sized for 40 kg of ballast	Two hoods used alternately. Decided by Amish, 2026-09-25
13	Battery	12 V 20 Ah sealed lead acid	Not drawn
14	Instruments	Steam gauge, four type K thermocouples and reader	Soil probes at 15 cm; not drawn
15	Safety kit	Personal CO alarm, fire extinguisher, gloves, face shield	Not drawn

#	Component	Choice	Notes
16	Feed and coil alarms	Coil outlet temperature alarm with a latching relay that cuts the pump until reset by hand, tank low-level switch	Decided by Amish, 2026-09-25; latching trip 2026-10-25 (STR-DEC-001 S4); not drawn



Figure 3. Exploded view with item numbers matching the BOM.

TRL 3 numbers

Requirement IDs are those of STR-PRS-001 (old R numbers in brackets). All values come from STR-CAL-001 v0.5 (`docs/04-calcs/01-sizing.md` , script `docs/04-calcs/sizing.py`). They are paper estimates; none is tested.

Table 2. Key results.

Quantity	Value	Requirement
Steam output	30 kg/h, firing 32.9 kW, firebox gas 519 °C	PR-003 (R3) met
Heat into the water	21.78 kW (coil 17.19 kW, bank 2.79 kW, economizer 1.80 kW)	
Fuel to steam efficiency	66.1 % at λ = 1.5 with the baffles (63.8 % to 68.3 % for F = 0.25 to 0.60; 62.5 to 66.9 % in the other sensitivities; 59.8 % without the baffles; STR-CAL-001, Tables 4, 6 and 6a)	PR-005 (R5) at risk: met only with the dampers set, inside its uncertainty
Stack temperature	257 °C	
Wood use	8.2 kg/h at 14.5 MJ/kg	
Steam per m² at 15 cm	15.6 kg/m² (front to 16.5 cm)	
Treatment rate, two hoods	1.88 m²/h at 15 cm; 3.41 m²/h at 5 cm	PR-002, PR-014 (R2) at risk (restated to about 1.9 and 3.4 m²/h)
Wood per m² at 15 cm	4.36 kg/m² (4.44 kg/m² with the dampers wide open)	PR-004 (R4) not met; restatement to 4.5 kg/m² proposed, awaiting Amish
Header pressure, normal	0.033 bar gauge	PR-006 (R6) met (0.1 bar at the header)
Coil inlet pressure, normal (at the bank inlet)	0.129 bar gauge	PR-013 (R6) met (0.15 bar at the coil inlet)
Water seal limit	0.084 bar gauge with the overflow	PR-006 (R6) open vent met by design
Water in heated section, flooded	7.32 L	PR-007 (R7) met
Steaming per fill	4.2 h (battery to confirm: about 100 % depth of discharge at an assumed 4 A pump)	PR-008 (R8) at risk
Relief valve orifice needed	6.2 mm at 30 kg/h; 13.4 mm for a dry-coil refeed flash (141 kg/h on a 30 s basis, which is open, DEC #20); 3/4 in valve rated 170 kg/h	PR-009 (R9) met on paper
Chimney outlet, hood outer skin	3.2 m; 23 °C	PR-010 (R10) met
Mass as towed, tank and seal drained	509 kg with both hoods aboard (643 kg full; 447 kg with the hoods on a second vehicle)	PR-011 (R11) not met
Overall width	1.48 m	PR-015 (R11) at risk
Parts cost	USD 2,720	PR-012 (R12): USD 520 over the USD 2,200 value-engineering target

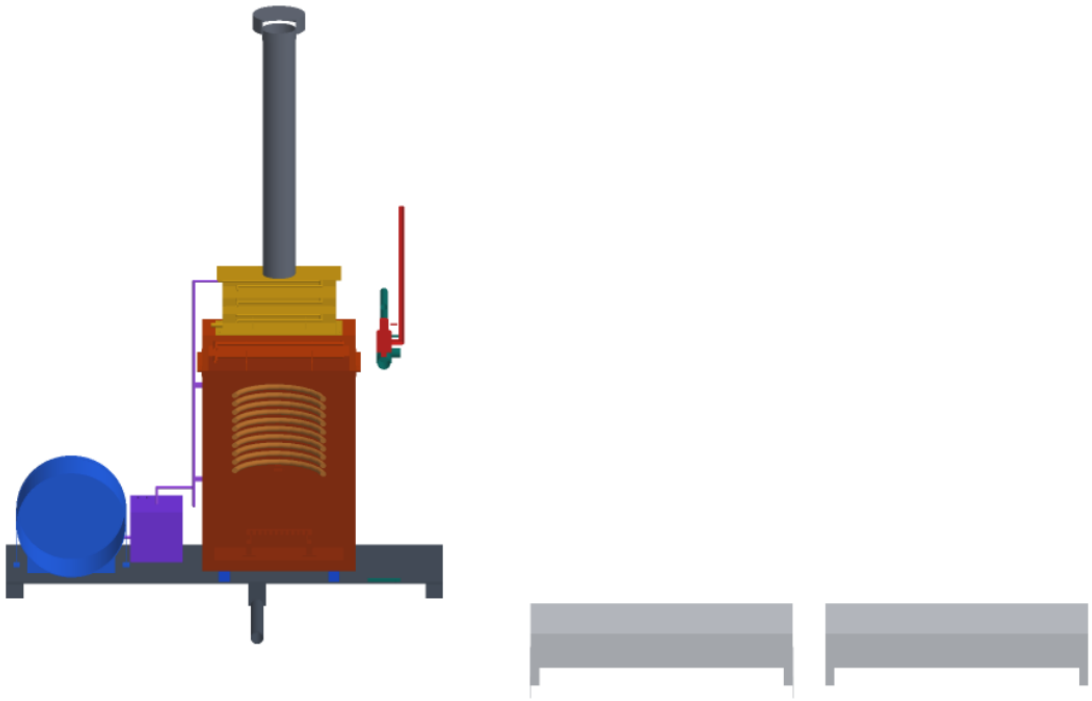
The main performance gap was heat transfer, and the kit 1.9.1 STEM check found it was larger than version 0.4 of the calculation showed: the bank and economizer coefficients it assumed (30 and 25 W/(m² K)) hold only if the gas crosses the tubes briskly, and in the open boxes it rises at about 0.1 m/s, where they are about 12. The gas-pass baffles fix that (about 22 and 21 W/(m² K) after fouling and edge bypass, 66.1 %), and the 3.2 m chimney gives the draft to push the gas through them with the fuel bed counted. The history: One coil in the firebox takes heat mostly by radiation, and without the bank the flue gas left the economizer at about 450 °C. Amish decided on 2026-09-25 to keep the 65 % target and study a convective evaporator bank with controlled air, and on 2026-10-02 to size it with primary and secondary dampers inside the 8 L limit of R7. The study in STR-CAL-001 v0.4, section 4, found that the dampers alone lift efficiency from 51 % to 61 %, and that a 6.4 m bank of smaller-bore tube with one coil turn fewer reaches about 69 % with 7.3 L of water if the gas side delivers what was assumed; with the baffles it reaches 66.1 %. Efficiency depends on the operator keeping the dampers set and on an unburned loss of 4 %; at 6 % it falls to 64.3 %, with the dampers wide open to 64.9 %. PR-004 needs about 72 %, so it stays short at 4.36 kg/m².

The second gap is steam per square meter. Soil behind the steam front sits at 100 °C, not 75 °C, and the front must go 1.5 cm past the working depth to hold 70 °C through the hold. In soils finer than sand, the steam cannot be pushed through 16.5 cm of soil at full flow without lifting a 31 kg hood, so real rates may be lower still. The hood handles are sized to carry 40 kg of ballast if the TRL 4 soil tests call for it.

SteamRoot: cutaway

github.com/BoujeeEnjinia1701/steamroot

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 4. Cutaway showing the helical coil above the fire bed inside the fiber-lined firebox, and the open-bottom hoods with their soil skirts.

Key design choices

- **Open-vented, never sealed.** The steam side is always open to the atmosphere through the water-seal pot and vent. There is no isolating valve between the coil and the vent, and the diverter can only send steam to the hose or the vent, never close both. Decided by Amish, 2026-09-25.
- **Monotube coil instead of a drum boiler.** A once-through coil and bank hold about 6 L of water, so a failure releases little stored energy. The cost is careful feed control: too little water dries the coil. Decided by Amish, 2026-09-25.
- **Large-bore stainless coil (25.4 mm, 316).** A 12.7 mm coil would need more than 2 bar at its inlet. The 25.4 mm coil keeps the header near 0.03 bar, though the inlet, with the bank ahead of the coil, is at about 0.13 bar. Stainless tolerates a dry-fire event far better than copper. Decided by Amish, 2026-09-25.
- **Hood steaming with two hoods used alternately.** Decided by Amish, 2026-09-25. One hose is moved between the hoods with a steam-rated coupling after steam is diverted to the vent. A second hose and a three-outlet diverter would avoid handling the hose but add at least $180. One hose: decided by Amish, 2026-09-25 (STR-DDR-002).
- **Tow drained, fill on site.** The machine is towed with the feed tank and seal pot drained (499 kg with both hoods aboard) and filled from a water supply at the site. Decided by Amish, 2026-09-25 (STR-DDR-002).
- **Relief valve sized for the refeed flash.** The 3/4 in, 15 psi valve is rated 170 kg/h, above the 138 kg/h a dry-coil refeed could produce. Decided by Amish, 2026-09-25 (STR-DDR-002).
- **Feed control.** Fixed pump rate with a coil outlet temperature alarm and a tank low-level alarm. Decided by Amish, 2026-09-25.
- **Fiber lining, not castable.** A 50 mm ceramic fiber lining keeps the firebox near 125 kg; castable refractory would make it about 406 kg and break R11. Fiber is less robust against loading damage. Fiber lining: decided by Amish, 2026-09-25 (STR-DDR-002).
- **Wood only.** Split wood or chips at 20 % moisture or drier, batch-fed by hand. No liquid fuel backup. Decided by Amish, 2026-09-25 (STR-DDR-002).

Safety

SteamRoot is the highest-risk concept in the portfolio. Every later stage must keep this section and extend it.

> **Safety: regulatory ruling first.** A written ruling from the local boiler authority (the Texas Department of Licensing and Regulation's boiler program; Amish's home jurisdiction is Texas) on this open-vented, fired monotube coil is an open prerequisite. No detail design, build or firing may happen before it, and its ruling overrides this document if it is stricter.
>
> **Safety: never sealed.** The steam side must always be open to the atmosphere through the water-seal vent. Never fit a valve, cap or plug that can isolate the coil or header from the vent. Keep the seal pot filled to its mark; an empty seal pot is an open vent, but an overfilled or frozen one raises the relief pressure. The overflow at the static water mark prevents overfilling. Drain the pot through its valve after every day of use and

whenever frost is forecast, and before lighting check that the pot is free of ice and filled to its mark (decided by Amish, 2026-10-02). Never plug, adjust or remove the relief valve. A sealed coil or tank over a fire can rupture violently.

Safety: certified relief valve. Fit only a certified, code-stamped steam safety valve sized for the full steam output. It is a second line of protection behind the open vent, not a replacement for it. Test it by its lever as the maker directs.

Safety: local boiler rules. Many jurisdictions regulate steam generators by pressure, heating surface, water volume or firing rate, and some regulate all fired steam equipment. The low-pressure design is intended to stay below common thresholds, but this is unconfirmed. Confirm with the local boiler inspector or authority before any build, and follow their ruling even if it is stricter than this document.

Safety: steam burns. Steam at 100 °C releases its latent heat on skin and causes deep burns within a second. Divert steam to the vent before lifting a hood or moving the hose coupling. In fine soils steam may jet from under the hood skirt, because the soil resists steam flow more than the hood's weight can hold (STR-CAL-001, section 2). Keep hands and feet clear of the skirt and the vent outlet. Wear heat-resistant gloves, a face shield, long sleeves and closed boots. Keep bystanders, children and animals at least 5 m away. Use a steam-rated hose with a whip check, and inspect it before every use.

Safety: dry coil. If the feed pump stops or the tank runs dry, the coil heats toward the firebox gas temperature (about 520 °C) while the fire keeps burning. Refeeding it could flash 1.2 kg of water or more (about 2 kg if the coil reaches 700 to 800 °C) at five times the design steam flow or more; the basis for sizing the vent path is still open (STR-DEC-001 #20). Never restart the feed into a hot, dry coil. Close the air damper, let the fire die and let the coil cool first. The coil outlet alarm and the tank low-level alarm exist to catch this early, and the coil alarm latches the pump off until it is reset by hand.

Safety: fire. Operate only on bare ground or gravel, clear of dry vegetation, with a fire extinguisher and water at hand. The spark arrestor must be fitted; the flue leaves at about 450 °C. Do not operate during fire bans or in high wind. Never leave a lit firebox unattended, and let it burn out fully before towing.

Safety: carbon monoxide. Wood fires produce carbon monoxide, which is odorless and deadly. The firebox and chimney must stay outdoors. Steam may be piped into a greenhouse or polytunnel, but the generator may not. The operator wears a personal CO alarm. Never refuel or tend the fire from downwind in the smoke.

Safety: hot surfaces and water quality. The firebox skin reaches about 100 °C, and the chimney, economizer, header, seal pot and hood inner skin reach burn temperatures. Use rain water or softened water to limit scale in the coil; scale raises tube temperature and can block the coil.

- **Soil biology.** Steaming kills beneficial organisms as well as pests, and the soil behind the steam front reaches 100 °C. Overheating soil can release manganese and ammonium that harm seedlings. Do not steam longer than needed.
- **Towing.** Tow only with the fire out and the tank and seal pot drained; the two hoods ride on a second vehicle (decided by Amish, 2026-10-02). Towing with a full tank puts the mass at about 633 kg, over the 500 kg limit in R11.

Open questions

These are tracked, with options and recommendations, in the design decisions register STR-DEC-001 (docs/06-design-decisions.md).

- Raised by the kit 1.9.1 upgrade (2026-10-25), proposed, awaiting Amish: the concept choice between this design (A) and a vented fire-tube kettle (B), which score within 10 % (#14); hood handling, since a 31 kg hood is beyond one person and borderline for two (S2, a wheel kit recommended); battery size (#16); three new product requirements (#17); bed preparation and hood hold-down for finer soils (#18); the render site scene (#19); the refeed flash basis (#20); a firebox skin limit (#21). Decided under the standing instruction of 2026-10-23: gas-pass baffles (S1), the 3.2 m chimney outlet (S3) and the latching pump trip (S4).
- Regulatory status of an open-vented, fired monotube coil in Amish's home jurisdiction, Texas: the written ruling is asked of the Texas Department of Licensing and Regulation's boiler program. Open prerequisite; gates all further work.
- First co-design partners (decided 2026-10-02): one market garden and one nursery within driving distance of Irving that already steam, solarize or chemically treat soil, found through Texas A&M AgriLife Extension's horticulture contacts. None is named or agreed yet.
- R5 and R4: decided by Amish, 2026-09-25, to study a convective evaporator bank with controlled primary and secondary air and keep the targets. Decided 2026-10-02: size the bank with primary and secondary air dampers so the heated water stays at 8 L or less without relaxing R7, reshape the firebox inside the same study, and plan to restate R4. Done on paper in STR-CAL-001 v0.4: R5 met at about 69 % with the dampers set, R7 met at 7.3 L, R6 met, R11 mass at risk at 499 kg. Proposed, awaiting Amish: restate R4 as 4.5 kg/m² or less with the dampers set.
- What soil permeability and skirt leakage do real beds show, and does the hood need ballast? This decides R1 and R2 and cannot be settled on paper. Decided 2026-10-02: ballast, skirt depth or steam rate are chosen from TRL 4 soil tests that measure leakage at the hood edge, and the hood handles are sized to take ballast weights.
- How much does starting soil moisture change steam demand? Wet soil needs more energy but conducts heat better.

Concept media: blueprint sheet, interactive 3D model. General arrangement: STR-DWG-002.